

SPECIALIZED MUCOSA

TONGUE AND TASTE BUDS

LECTURE LEARNING OUTCOMES: SPECIALIZED MUCOSA

LLO

By the end of this Lecture , students should be able to:

1.

Discuss the macro and micro anatomy of dorsal surface of tongue

2.

Explain anatomy and microscopy of different lingual papillae

3.

Discuss taste buds and taste sensation

Essential reading:

- **Tencate's Oral histology**
- **James Avery textbook**

Based upon Primary functional criteria,

1. Masticatory Mucosa (25%) : Gingiva, Hard palate

2. Lining Mucosa (covers 60% of total area): Lip, cheek,
vestibular fornix, alveolar mucosa, floor of mouth and soft palate

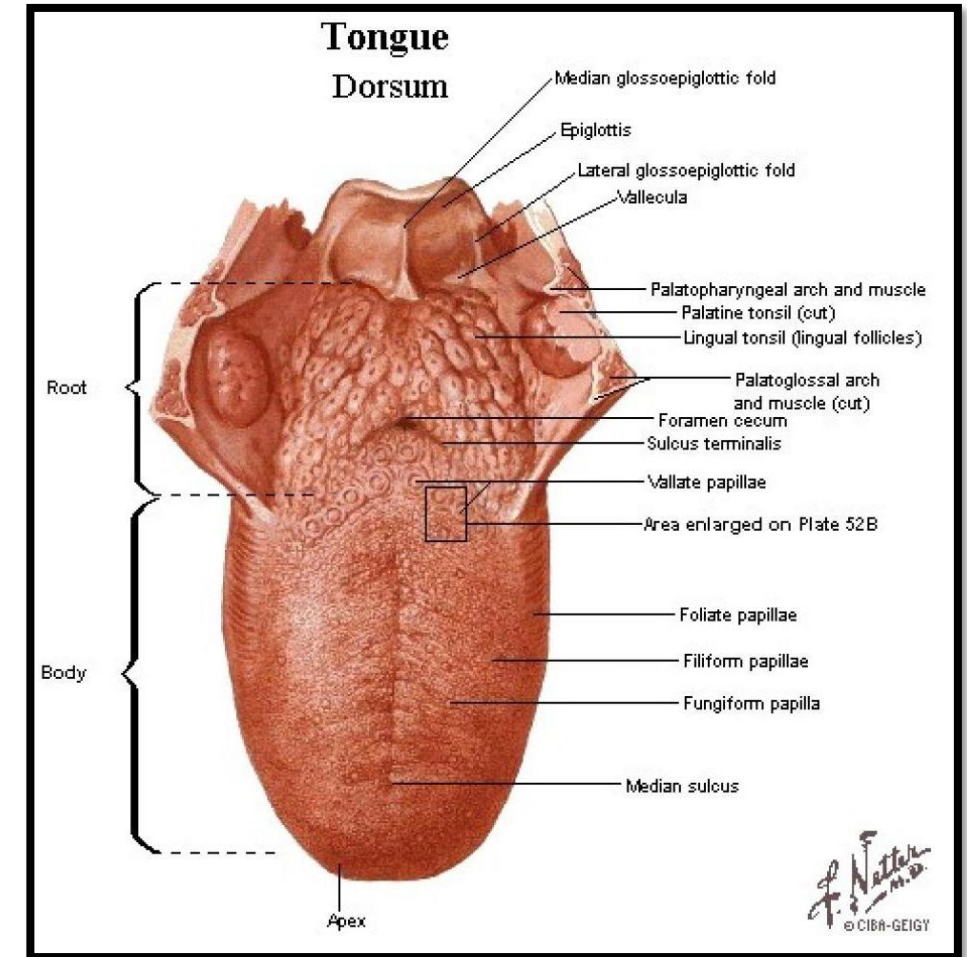
3. Specialized (sensory) Mucosa (15%): Dorsum of the tongue and
taste buds)

TONGUE

Tongue is a mobile muscular organ.

Functions:

- Taste perception.
- Speech (articulation).
- Mastication.
- Deglutition.

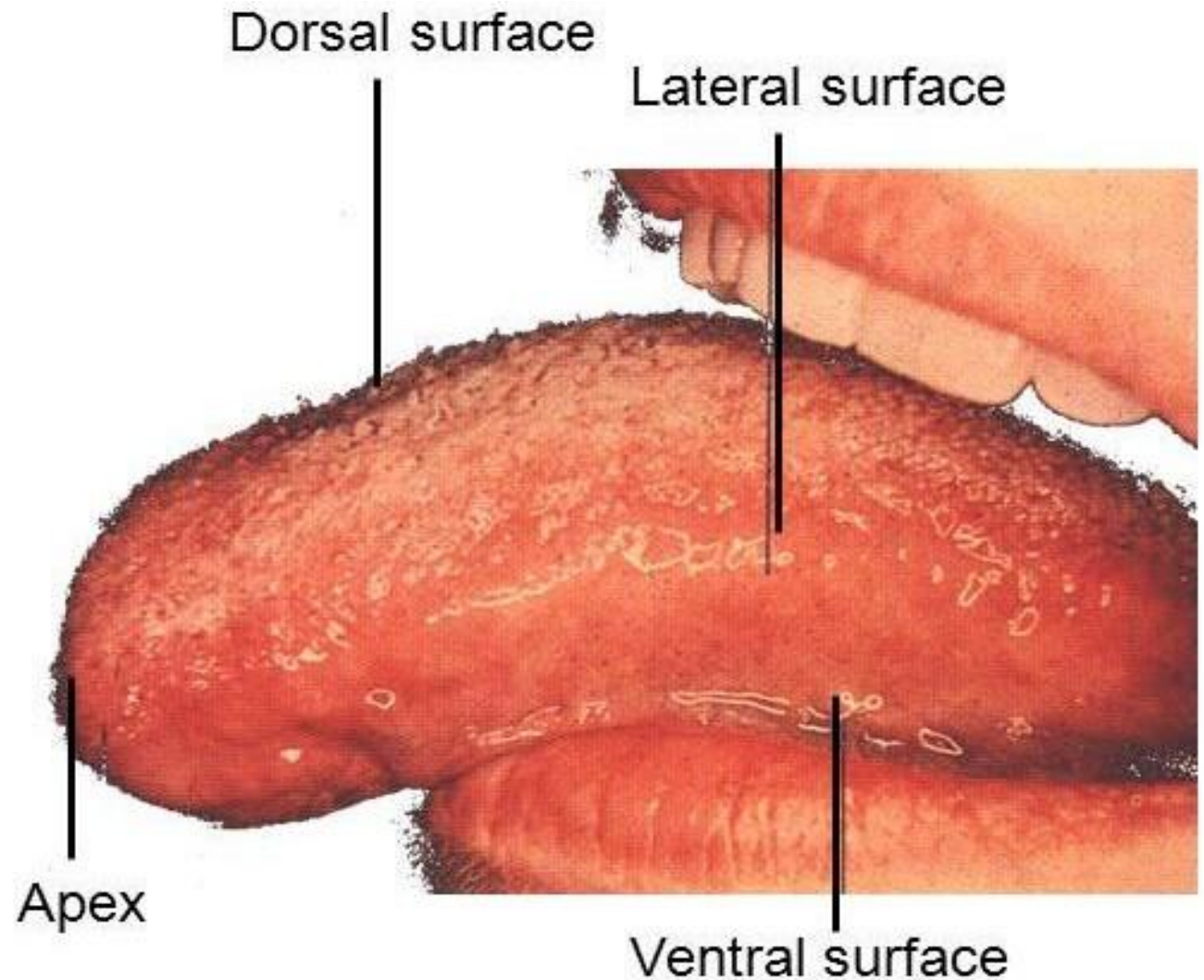


It has **two surfaces**:

- Dorsal
- Ventral

It has **two borders**:

- Right lateral
- Left lateral

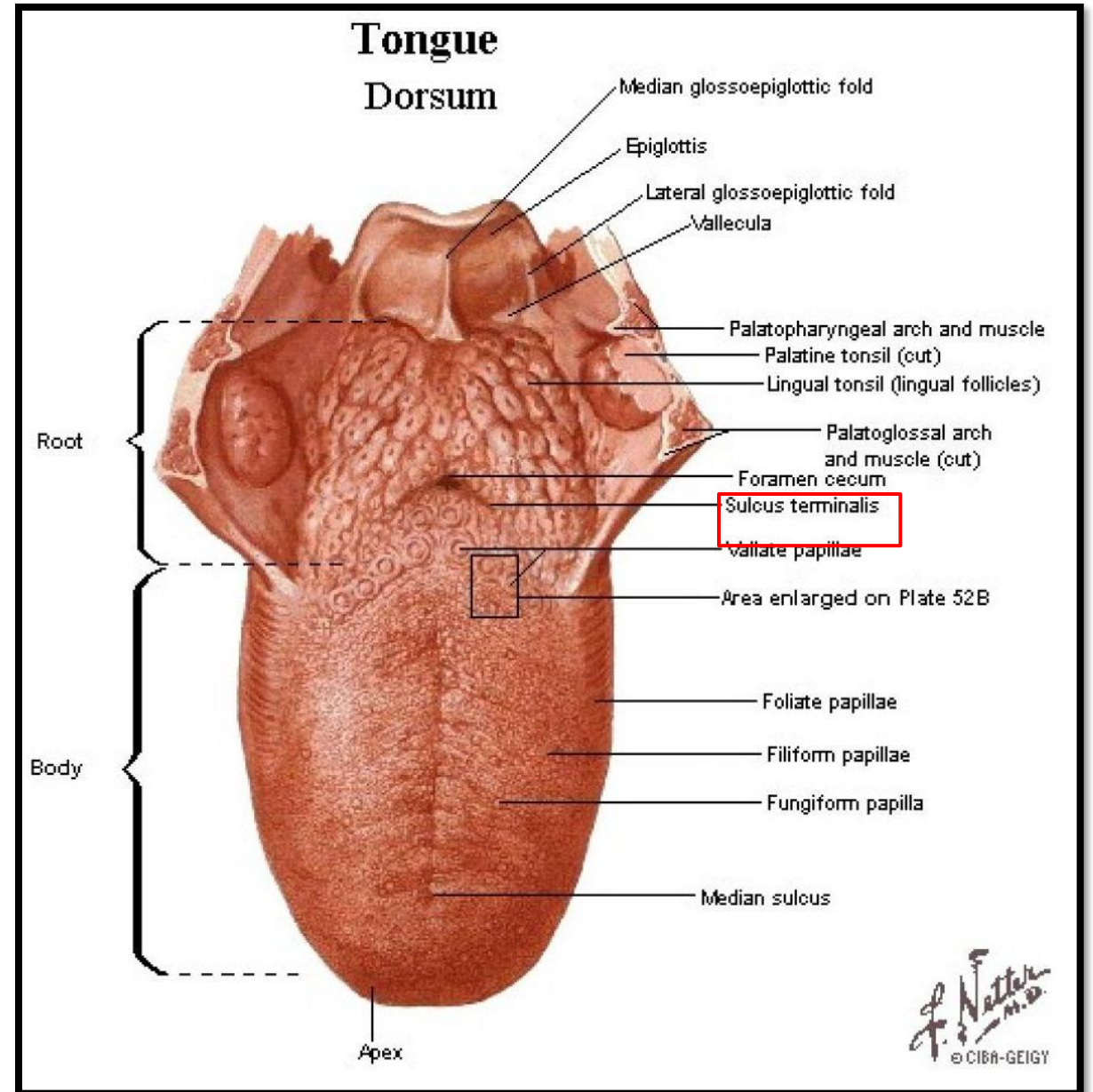


Dorsal Surface

- Convex.

Sulcus Terminalis:

- An inverted V-shaped groove on the dorsum of the tongue.
- Divides the dorsum of tongue into anterior $2/3^{\text{rd}}$ and posterior $1/3^{\text{rd}}$.
- **Anterior $2/3^{\text{rd}}$ – Body** of tongue
[Oral part]
- **Posterior $1/3^{\text{rd}}$ – Root** of tongue
[Pharyngeal part]



NERVE SUPPLY OF TONGUE

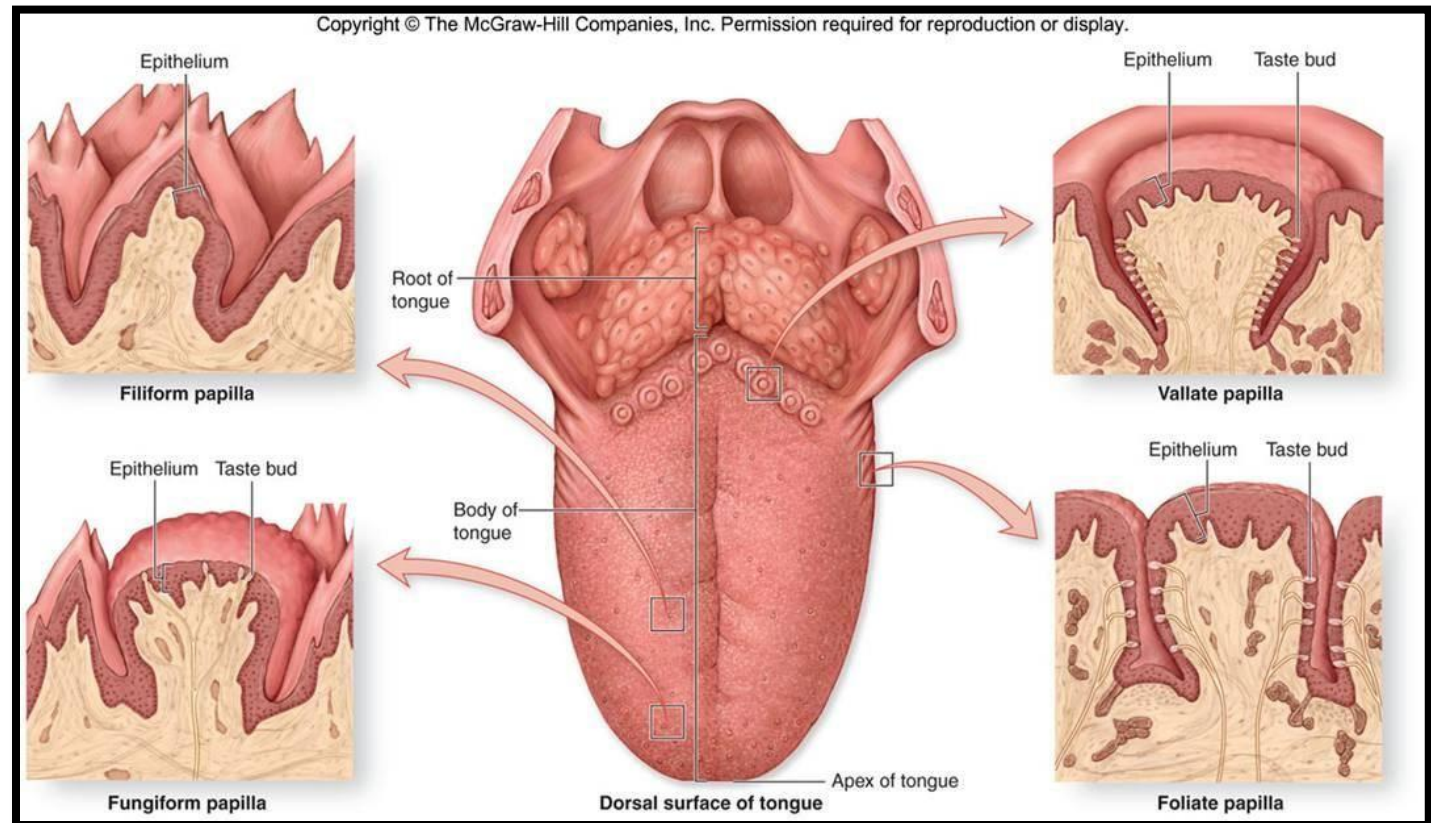
- **Anterior 2/3** $\longrightarrow$ comes from the chorda tympani branch from the facial nerve.
- **Posterior 1/3** $\longrightarrow$ glossopharyngeal n. •
- **Epiglottis & pharynx** $\longrightarrow$ vagus nerve

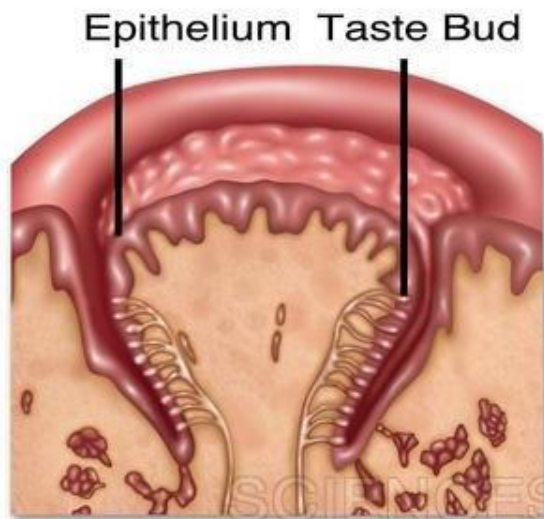
Lingual papillae:

- These are projections of lamina propria covered with epithelium.

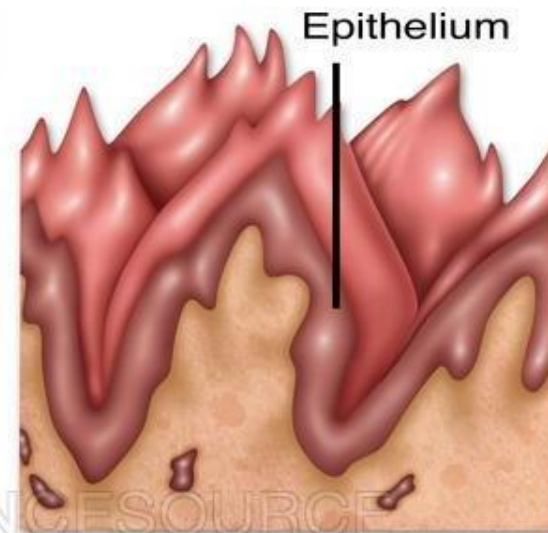
Types- 4

1. Vallate papillae.
2. Filiform papillae.
3. Fungiform papillae.
4. Foliate papillae.

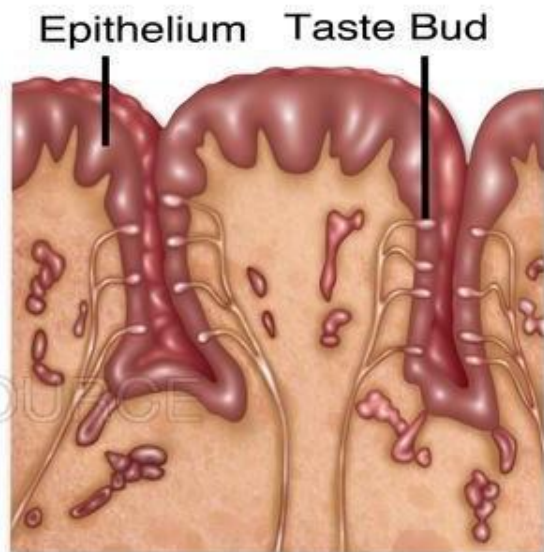




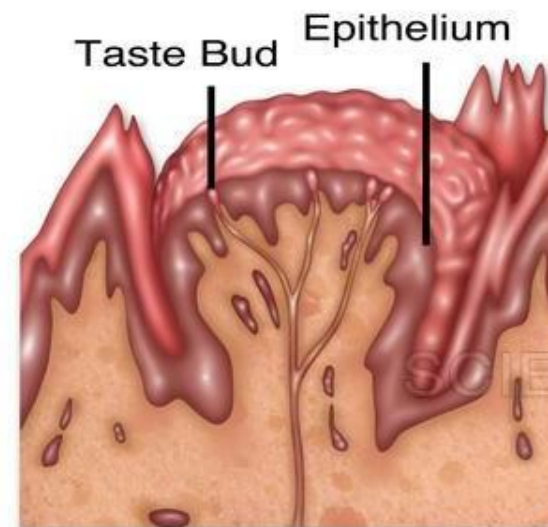
Vallate Papilla



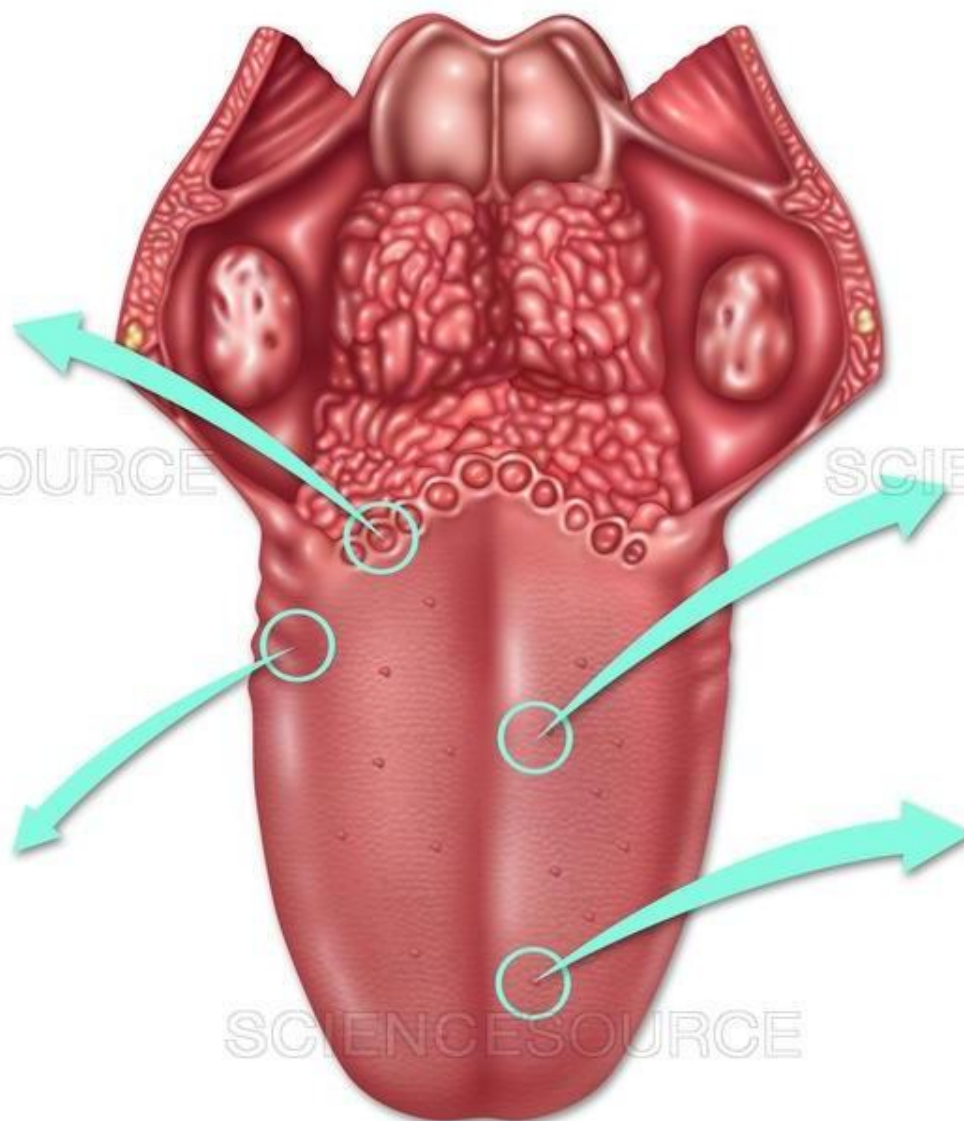
Filiform Papilla



Foliate Papilla



Fungiform Papilla



Tongue



Filiform Papillae

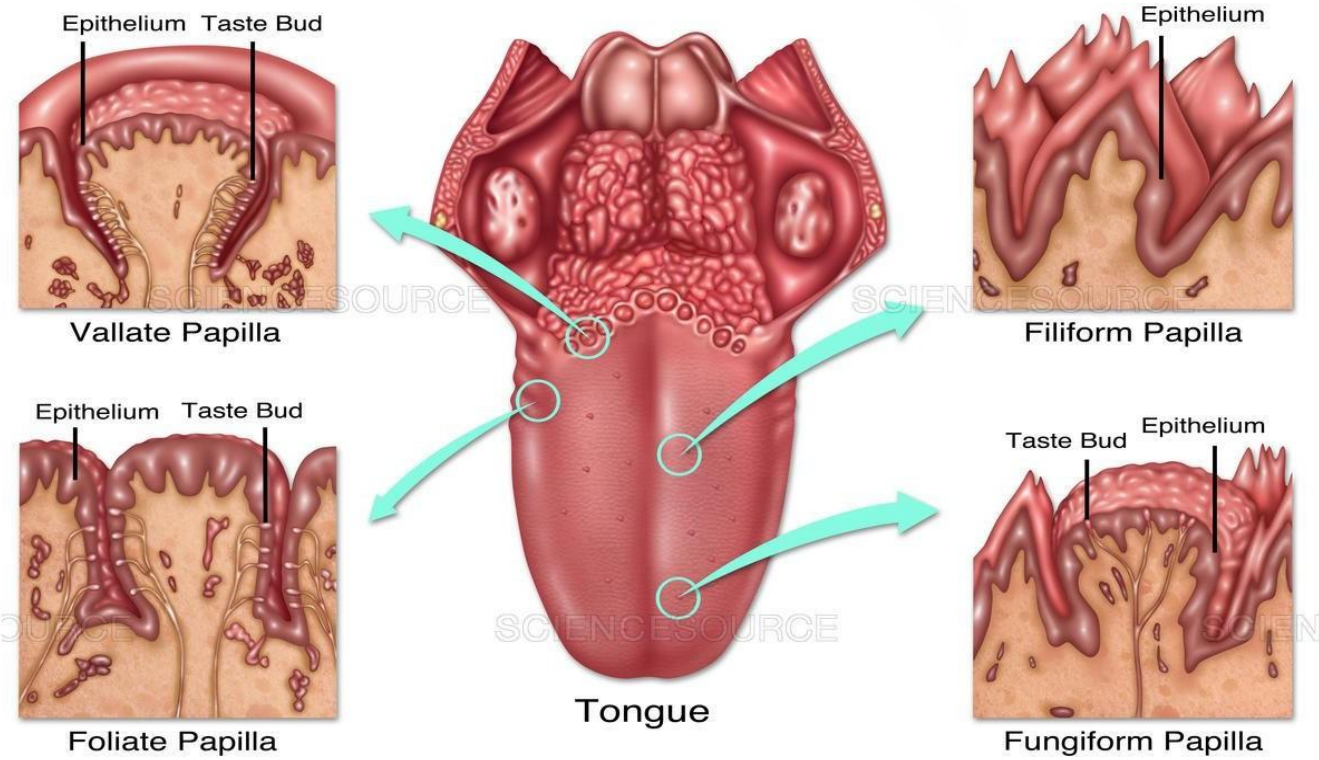
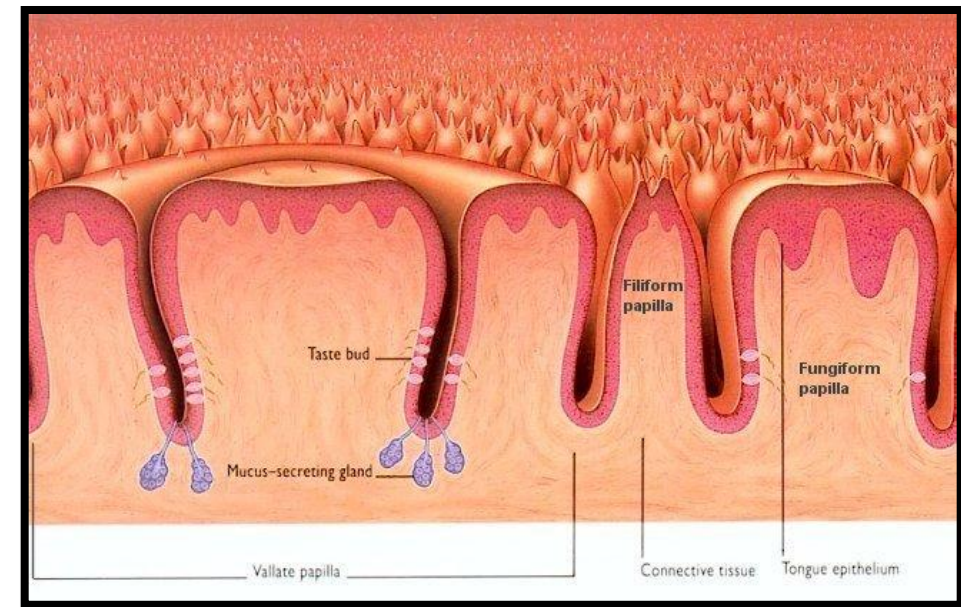
- Most numerous.

Shape- finger like with sharply pointed tips.

Location- all over the dorsal surface of oral part of tongue. (ant 2/3 rd)

No taste buds.

Covered by keratinized epithelium



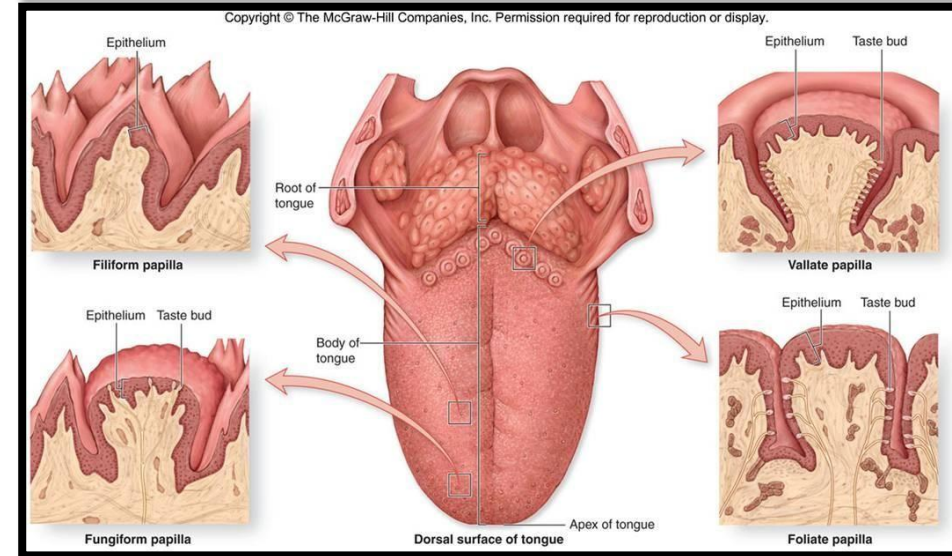
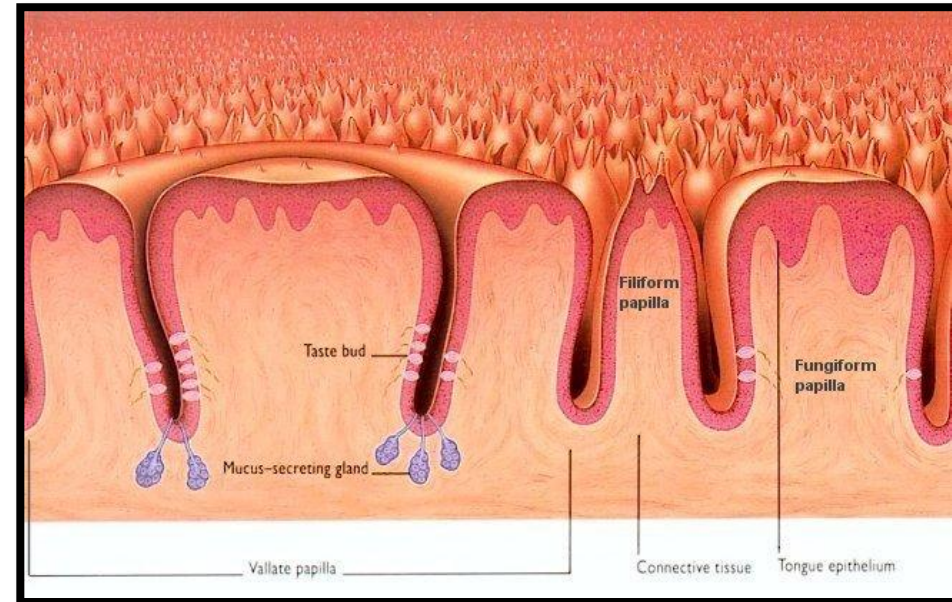
Fungiform Papillae

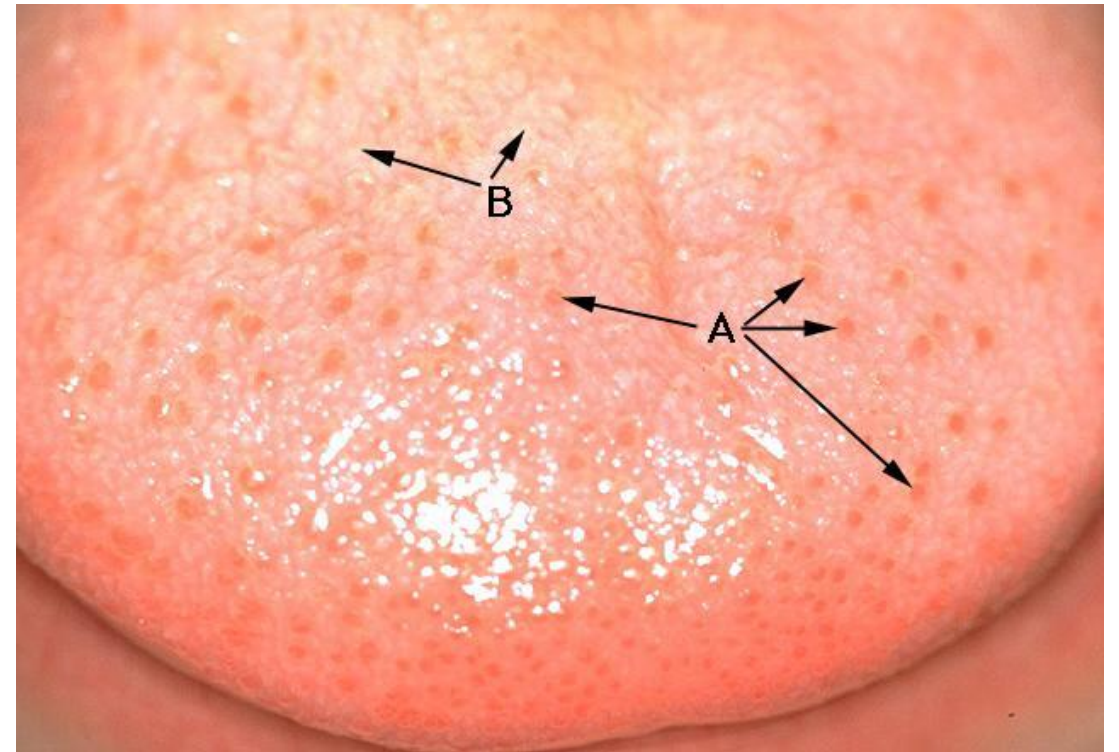
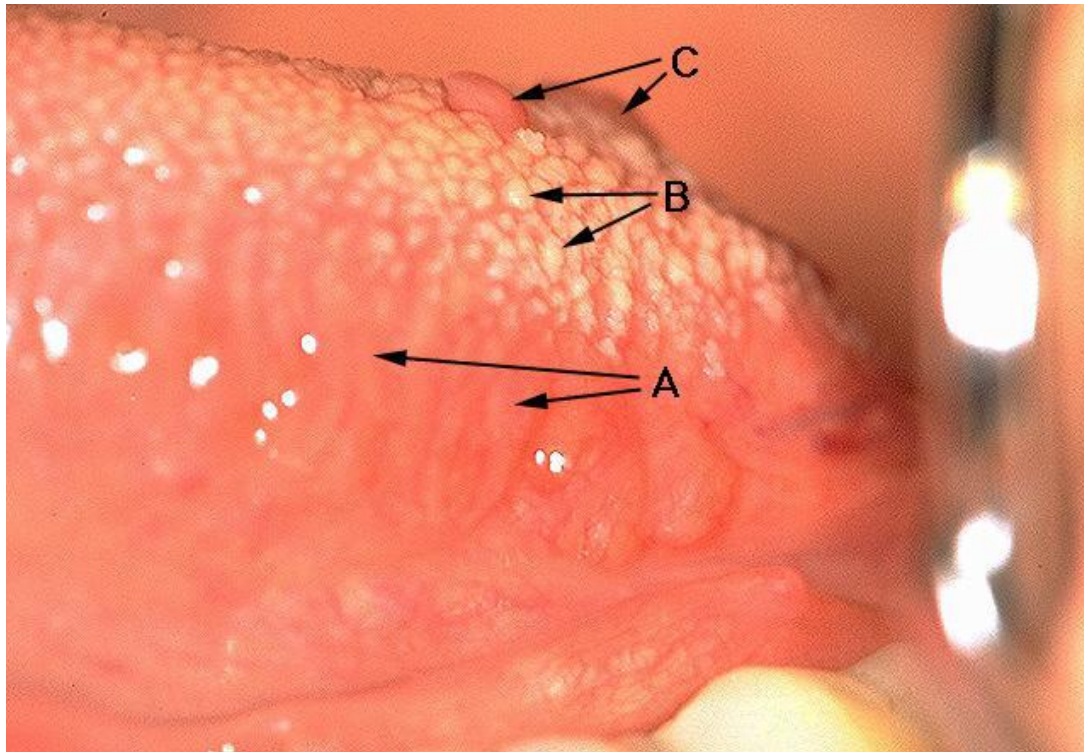
- ~ 1mm in diameter. Less in number as compared with filiform papillae

Shape- mushroom shaped (rounded head and narrower base).

Location- mostly at the **apex** and **margins** of the tongue.

- Visible as discrete pink pinheads.
- **Taste buds** are present at superior surface.
- Covered with thin **non keratinized epithelium**





A - fungiform papillae

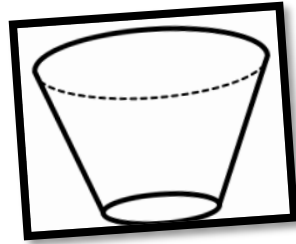
B - keratinized filiform papillae

Vallate Papillae [Circumvallate Papillae]-

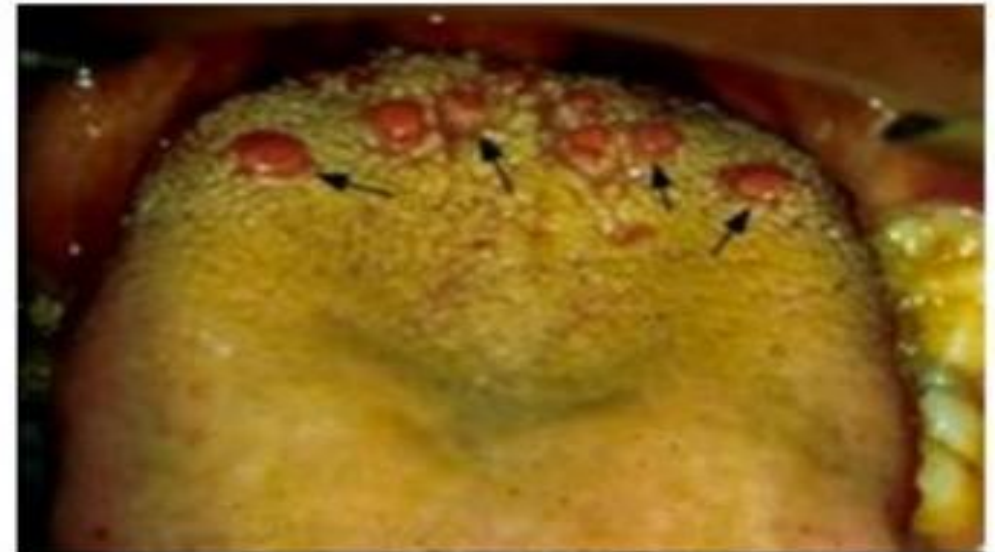
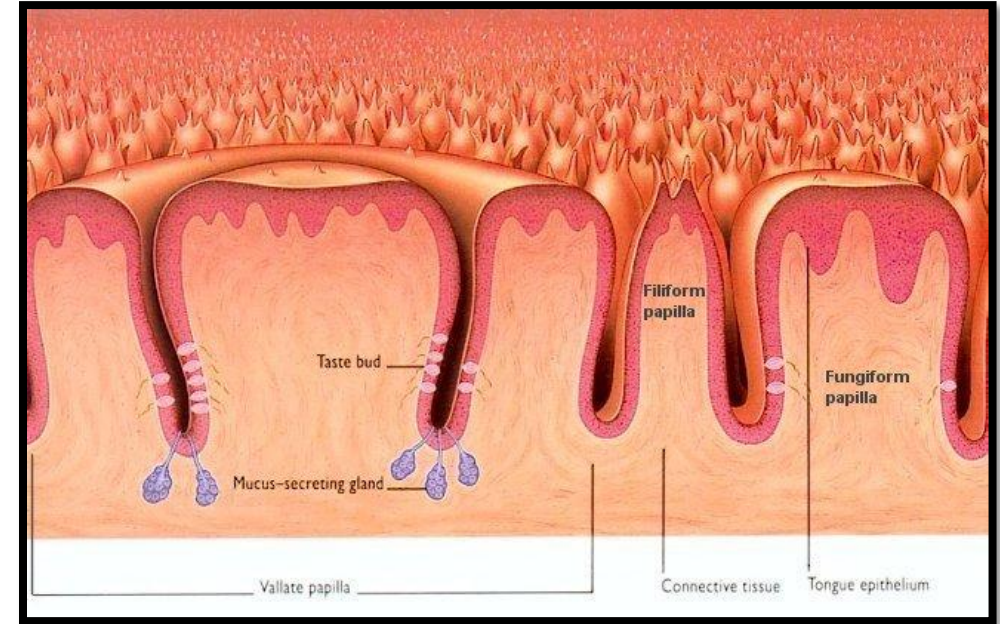
- **Largest** (1-2 mm in diameter). 8-12 in Number

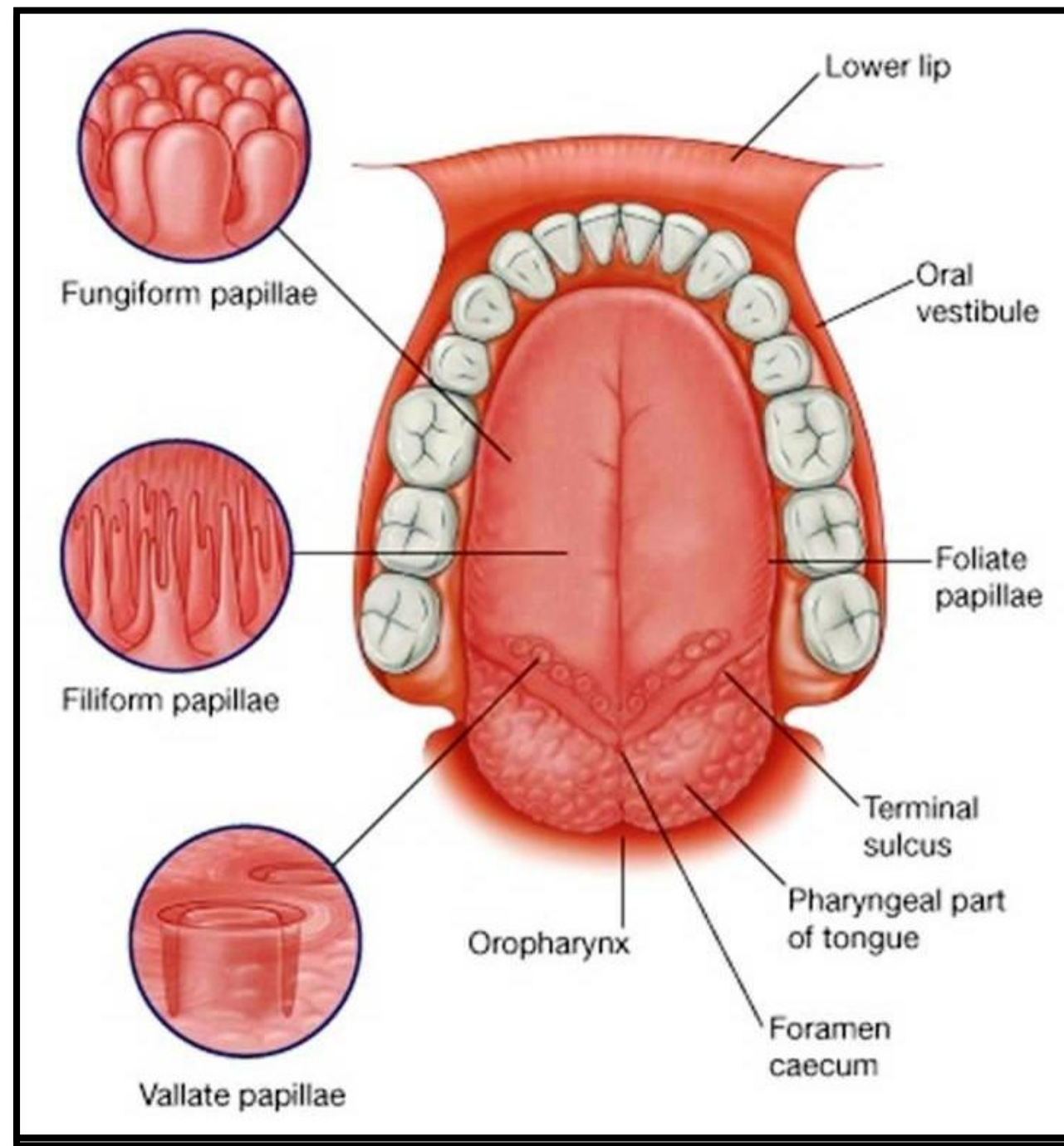
Location- In front of sulcus terminalis in a 'V' shaped row

Shape- Truncated cone.



- Each papilla is surrounded by a circular sulcus or trench.
- Circular sulcus is bounded on its periphery by a wall (vallum).
- Ducts of several salivary glands open in the circular sulcus. (**VON-EBNER MINOR SALIVARY GLAND**)
- **Taste buds are present at lateral side** in the papilla and its vallum.
- Covered with thin **non keratinized epithelium**.





HISTOLOGY OF TONGUE

Mucosa

- **stratified squamous** epithelium on both surfaces.
- The dorsal surface forms **papillae**

Submucosa

- Has **minor salivary** glands that secrete both mucous and serous secretions.

Muscles

- Core of **skeletal striated** muscles running in all directions

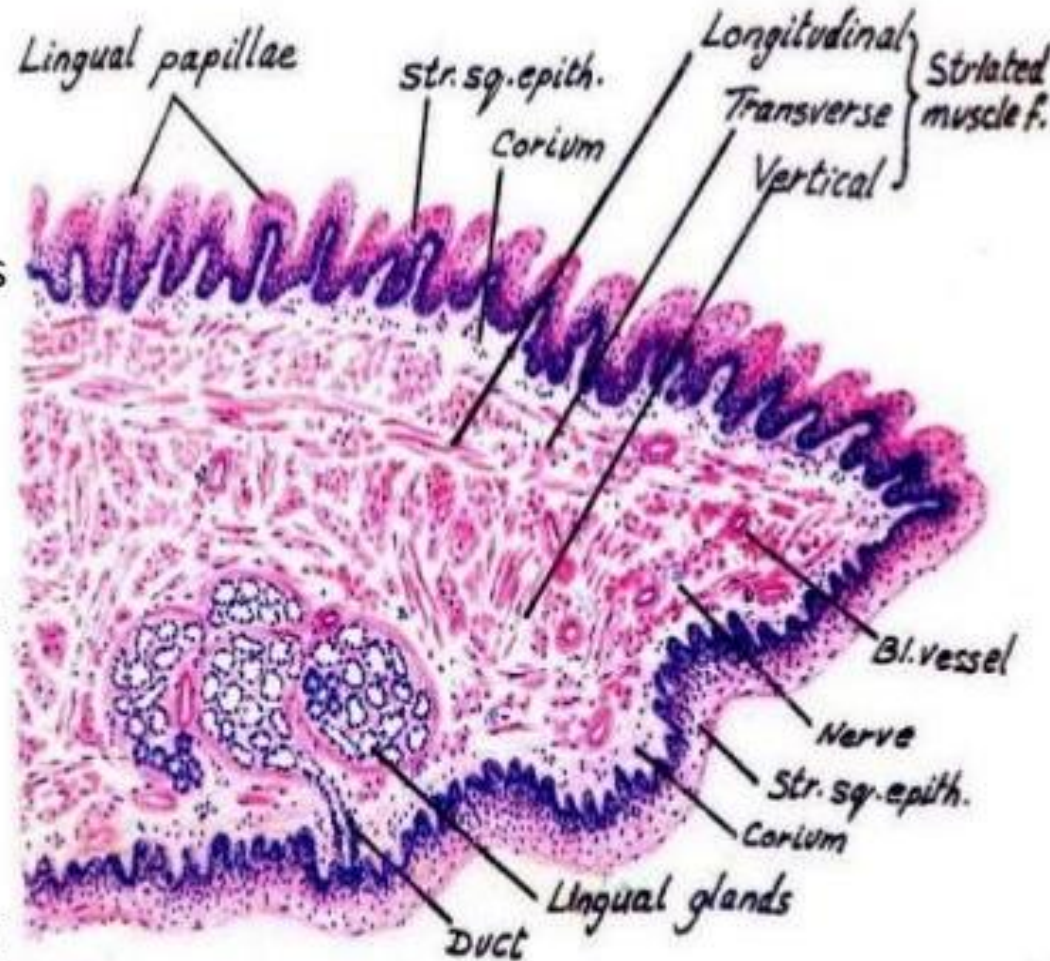
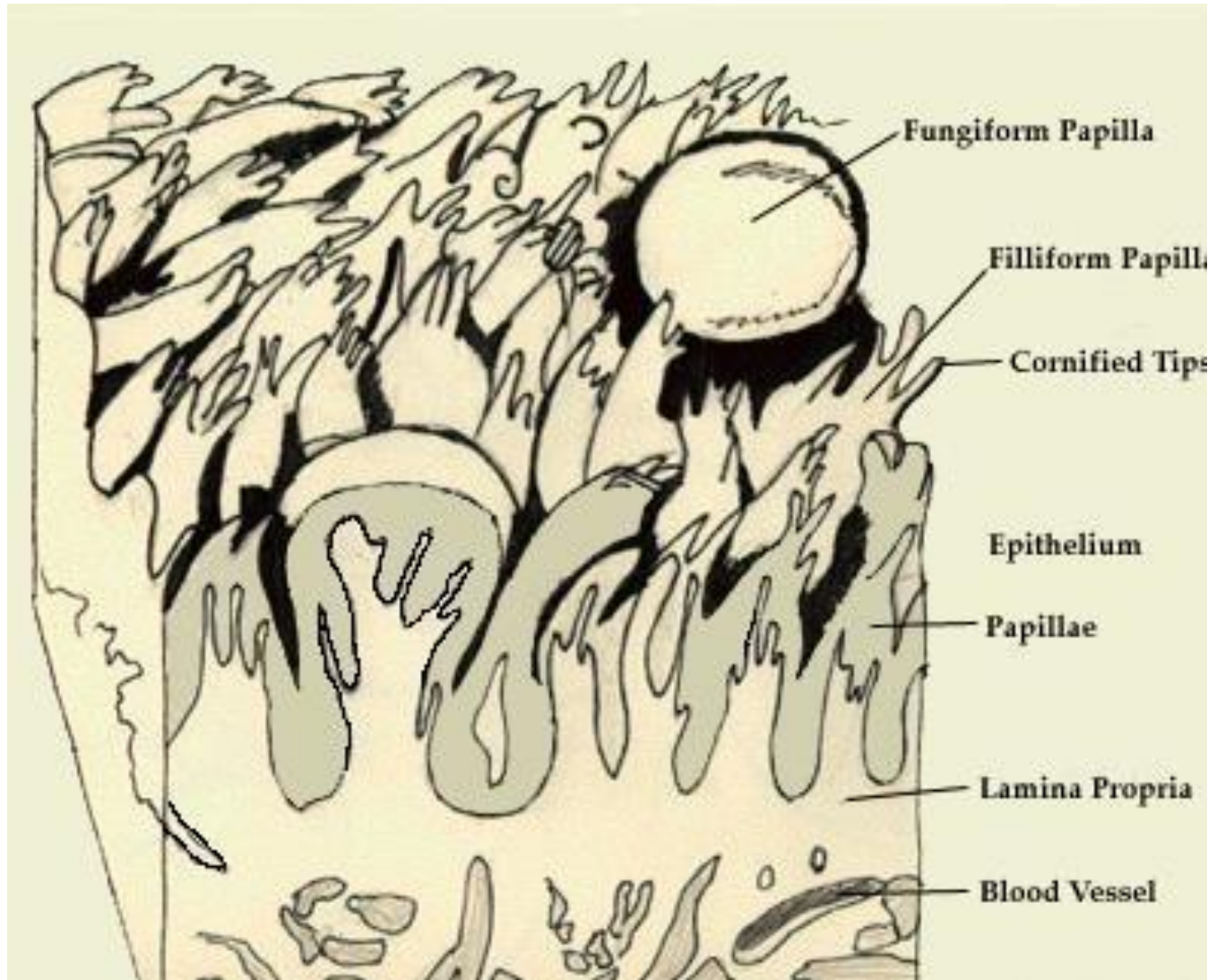
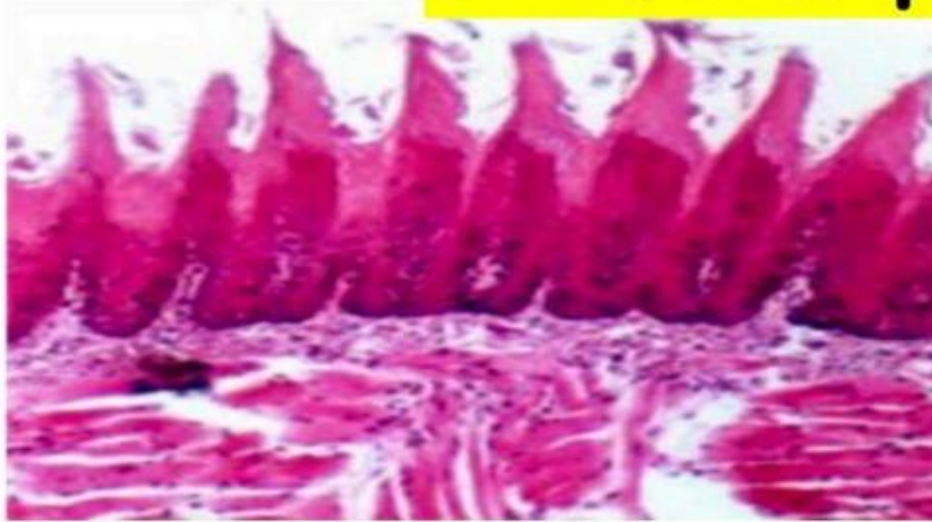


Fig. 3 Histology of the Tongue: Pearson Education 2013



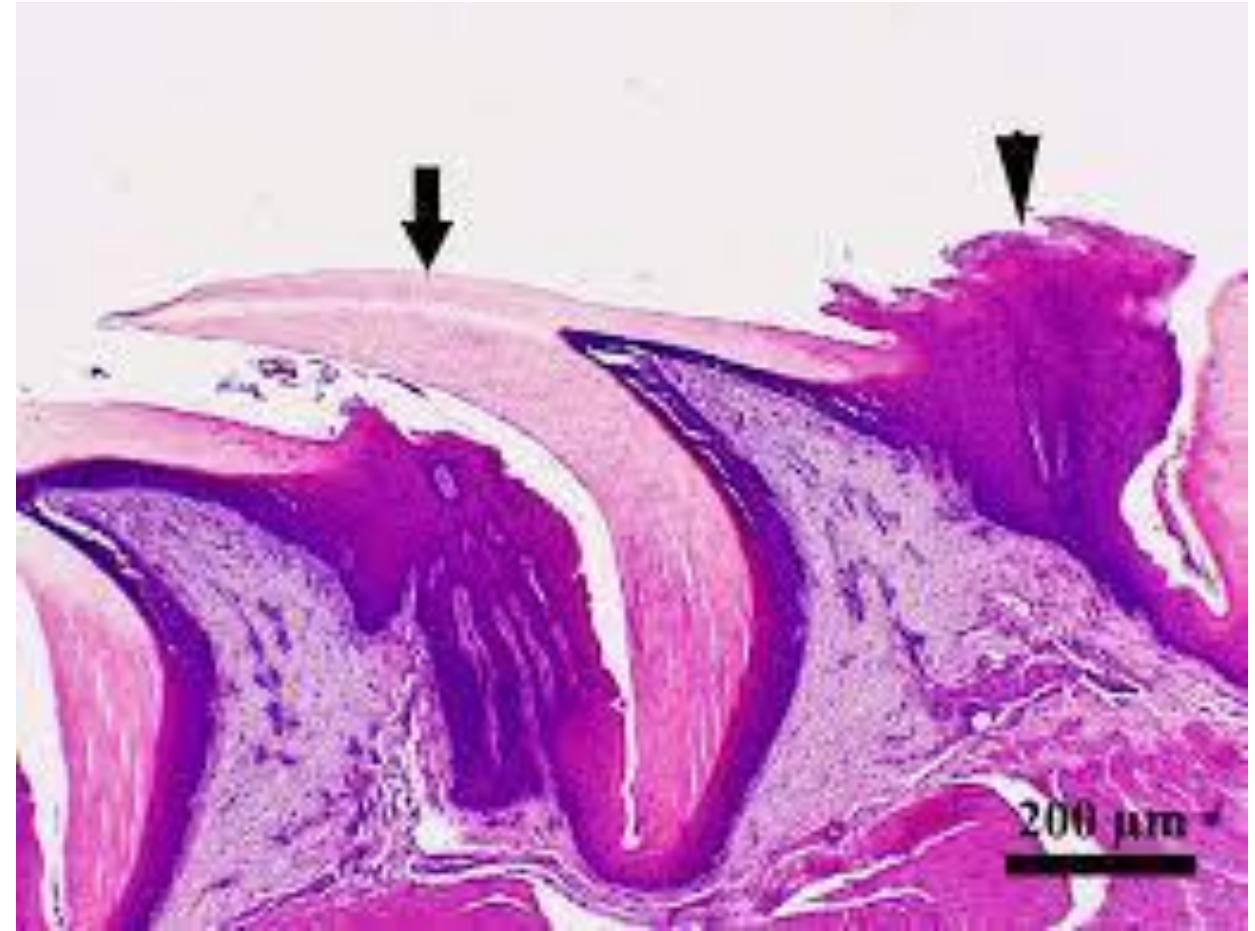
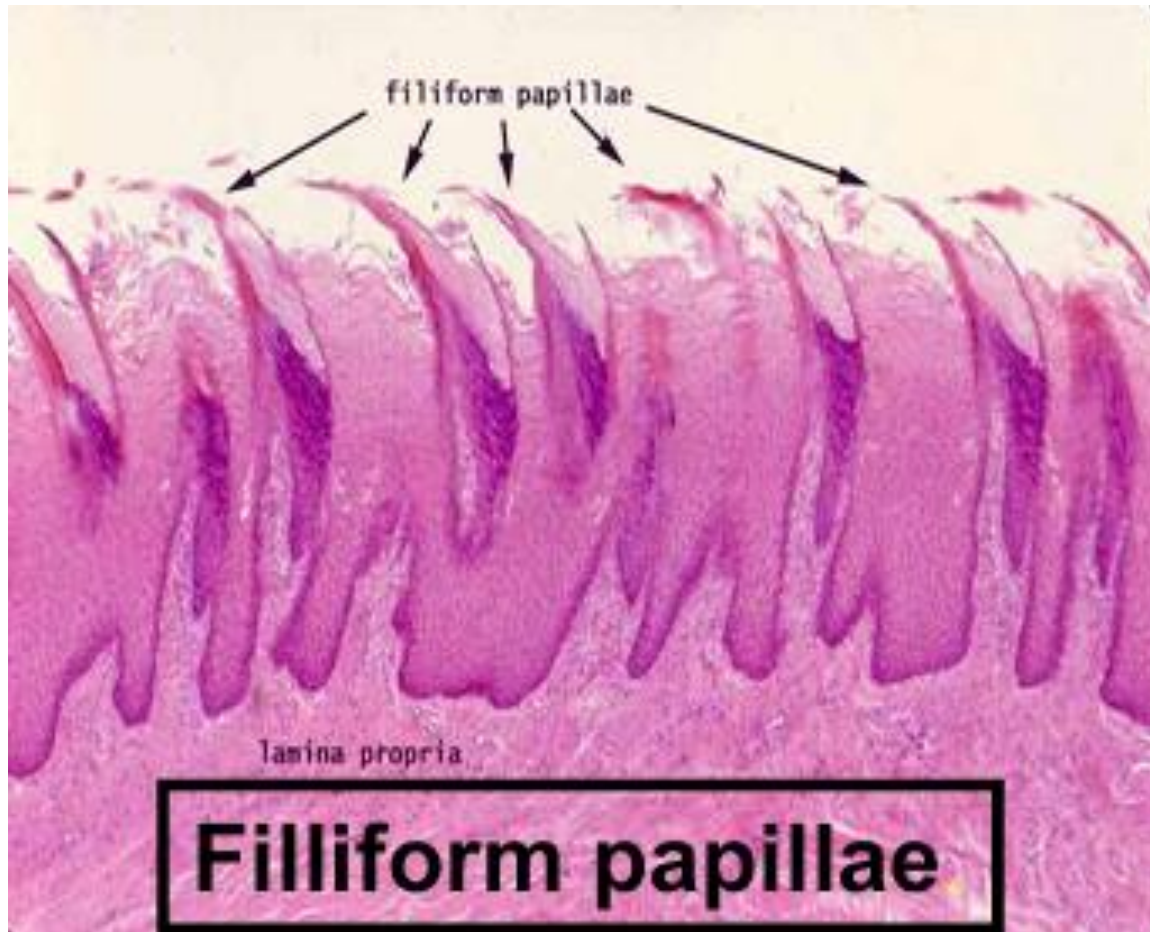
Cut-away section of tongue to show three-dimensional view of papillae and underlying c.t. and muscle.

Filiform papillae



- They are scattered over the surface of the anterior 2/3 of the tongue.
- They are slender , thread-like papillae, extend from 2-3mm high from the surface.
- **Function:**
- They are covered by keratinized epithelium.(act as protective layer for tongue)& give the tongue a roughened surface.
- Their pointed tips are directed towards the throat and so they assist in moving the food in that direction.

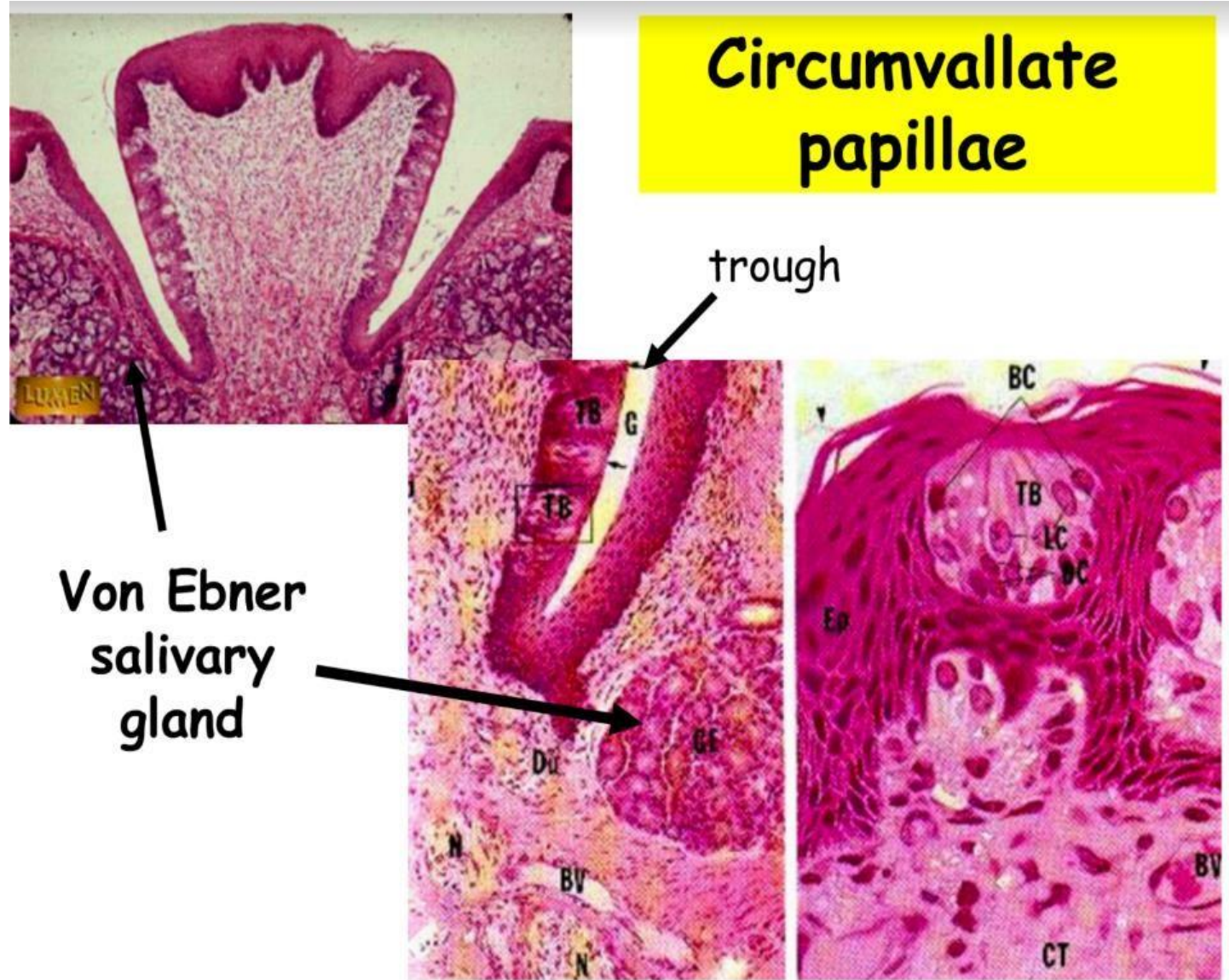
Filiform Papillae

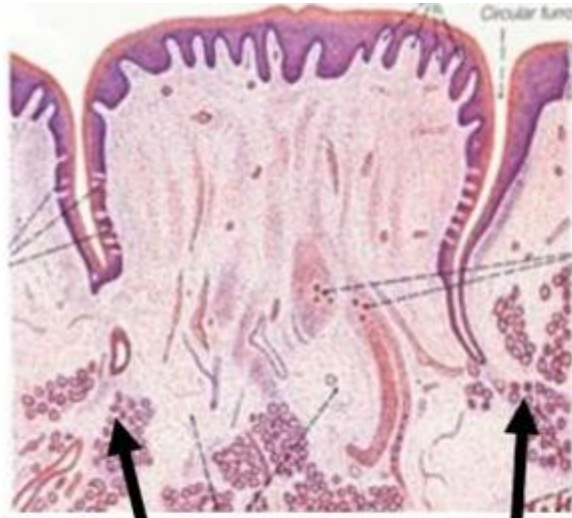


Fungiform Papillae

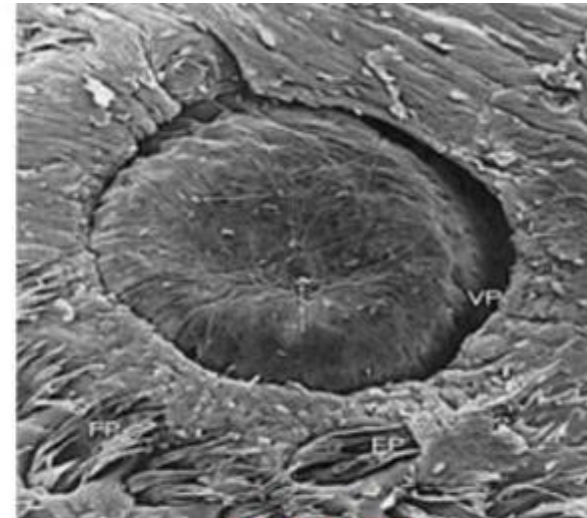
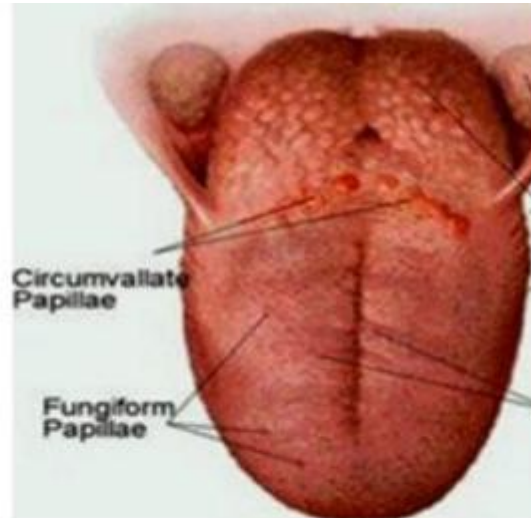


Circumvallate Papillae

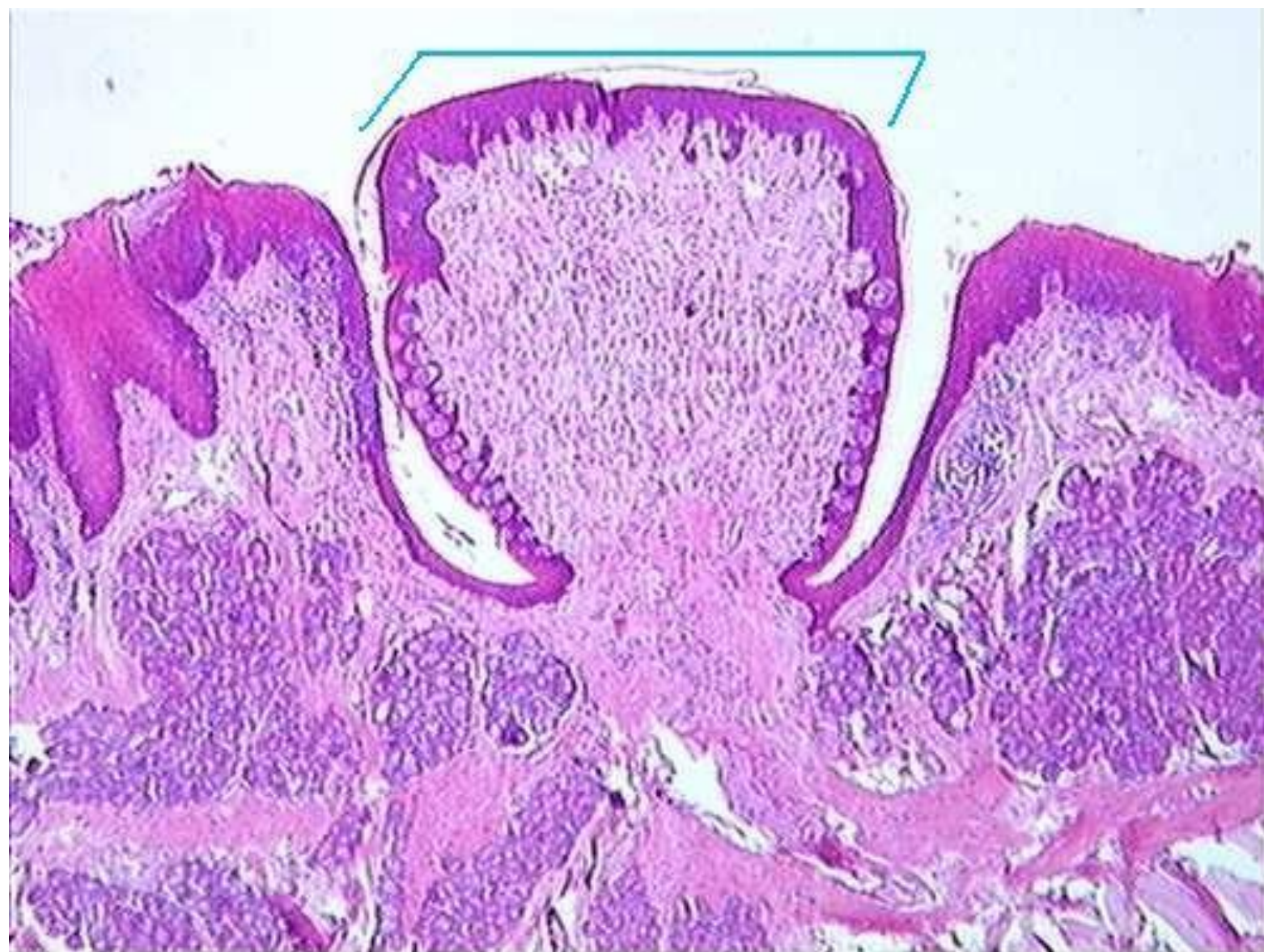


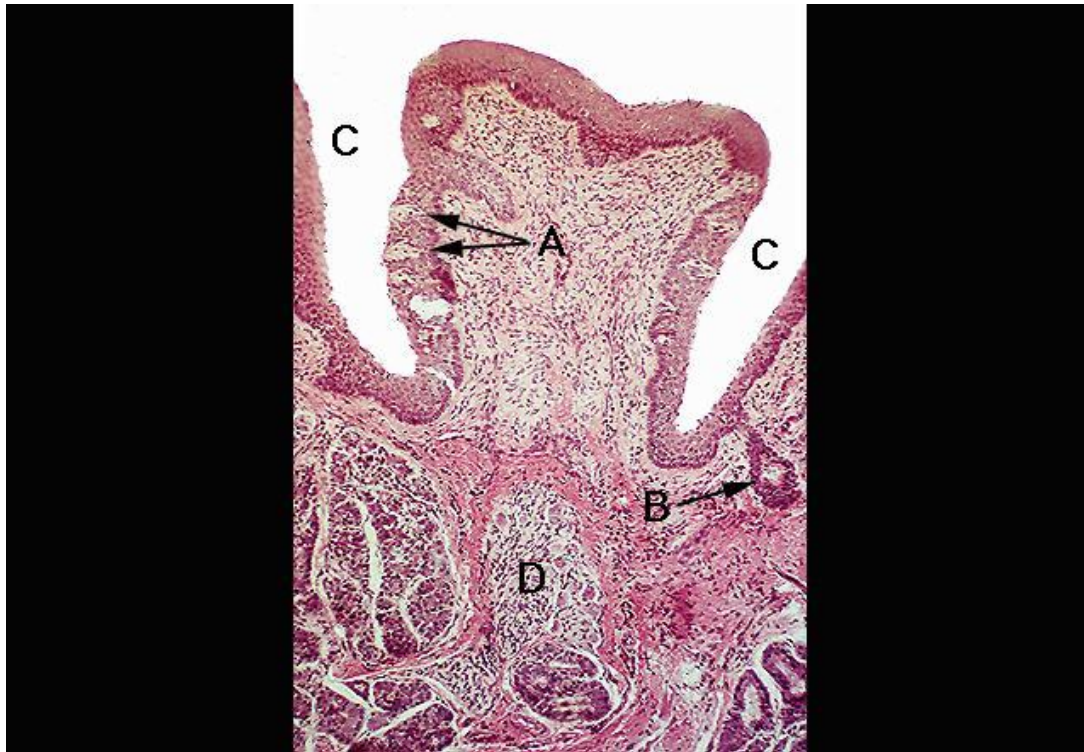


Von Ebner salivary glands



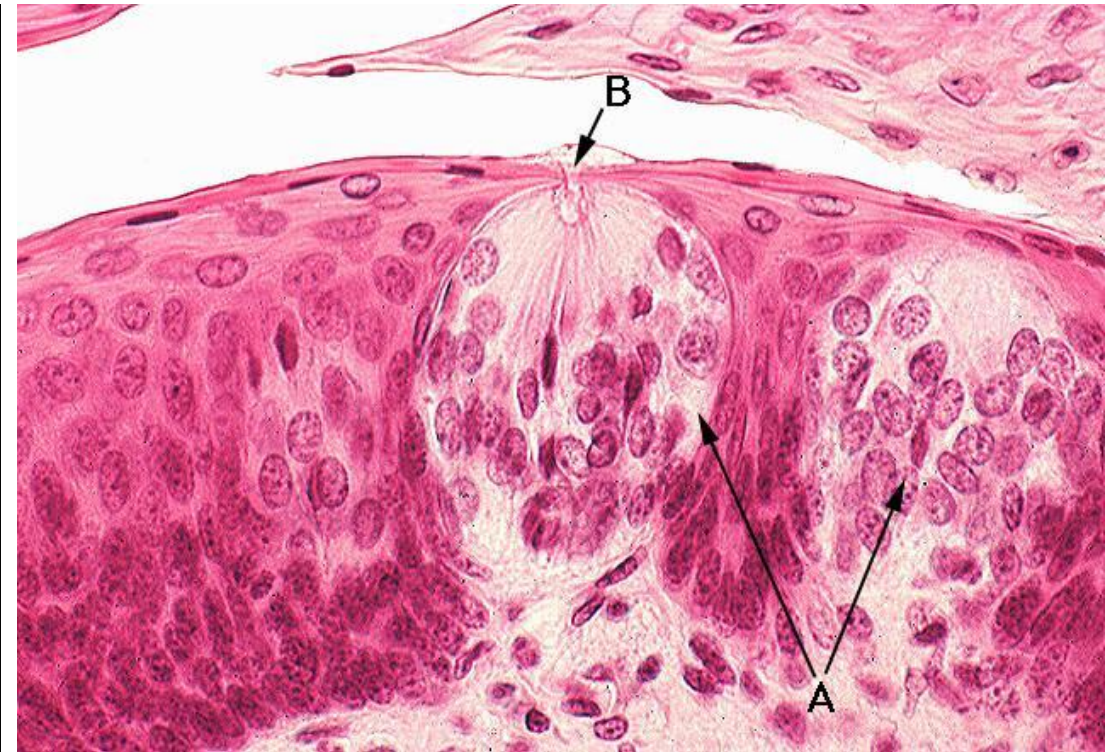
"SEM"





A - taste buds
B - duct of gland of von Ebner

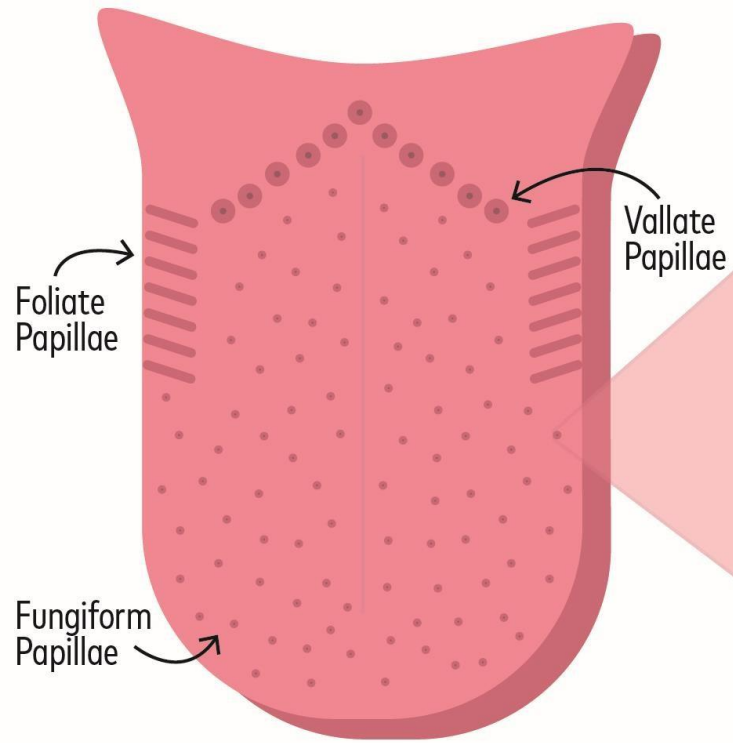
C - moat around circumvallate papilla
D - nerve



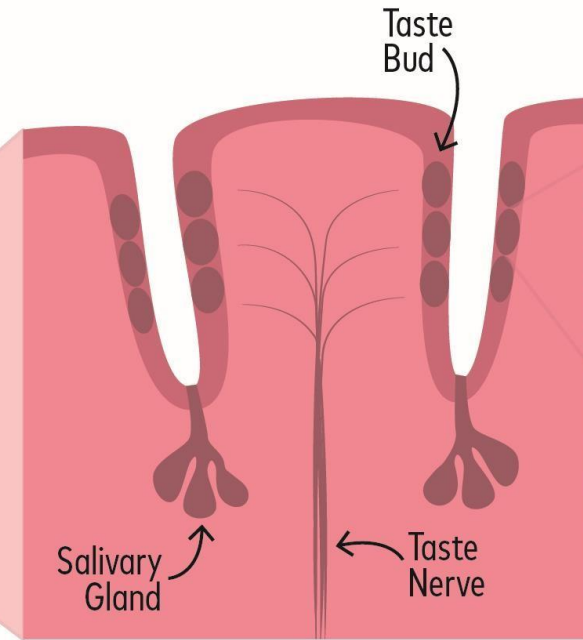
A - taste buds

B - taste canal

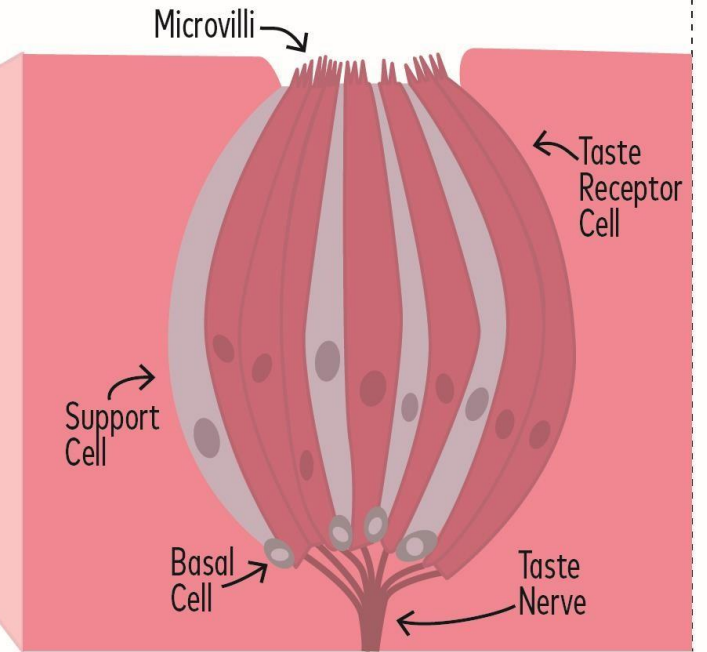
Tongue



Papilla

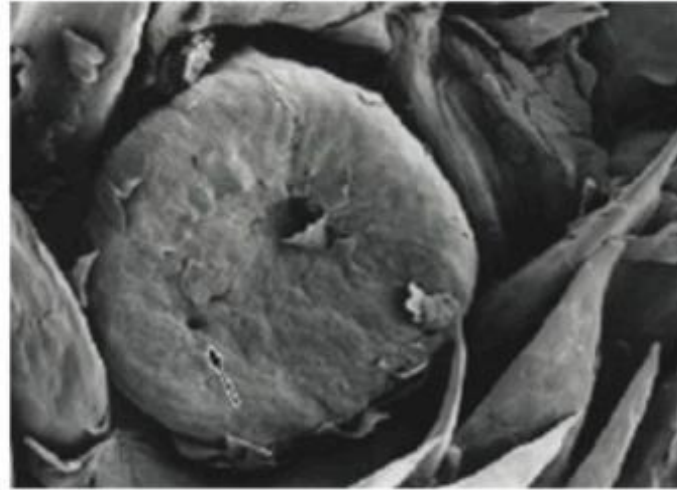


Taste Bud



Filiform	Fungiform	Circumvallate	Foliate
Most numerous Anterior part of the tongue	Present between Filiform papillae mostly at the tip and lateral surface	Present in the anterior 2/3 rd of the tongue near the sulcus terminalis	Present on lateral surface of the tongue
Appears as tiny conical projections	Mushroom shaped papillae	- Presence of deep surrounding groove - Von Ebner glands in Submucosa that wash out the grooves around the papillae for new taste perception	- Present as parallel ridges - Von Ebner glands in Submucosa
Always Keratinized	Nonkeratinized but can be keratinized	Nonkeratinized	Nonkeratinized
Absence of taste buds	Contains taste buds on superior surface	Contains taste buds in lateral wall	Contains taste buds in lateral wall

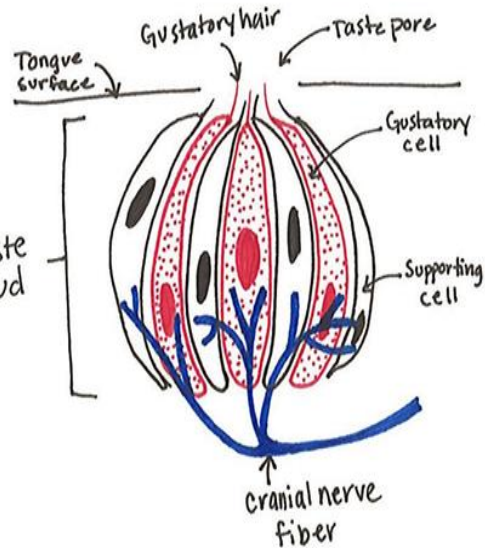
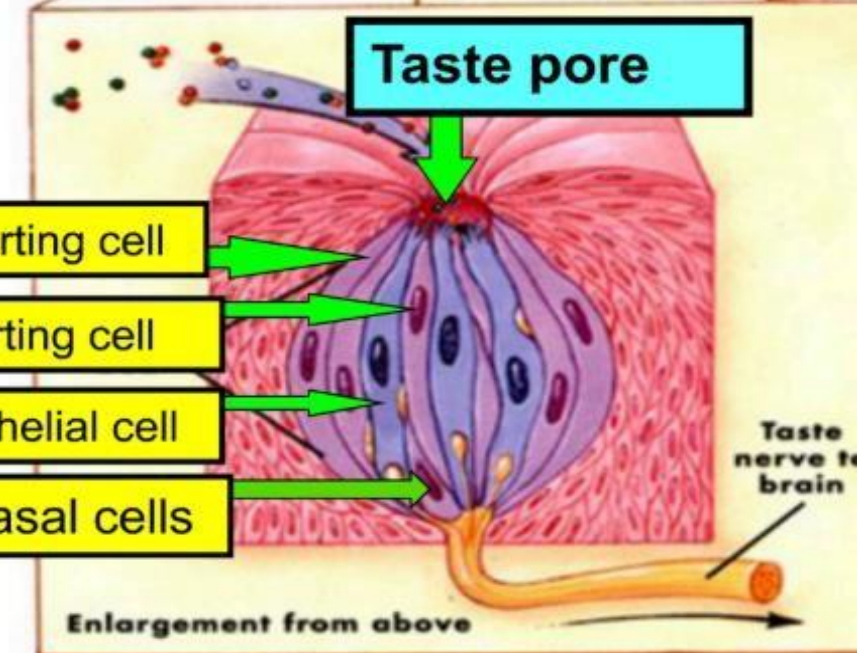
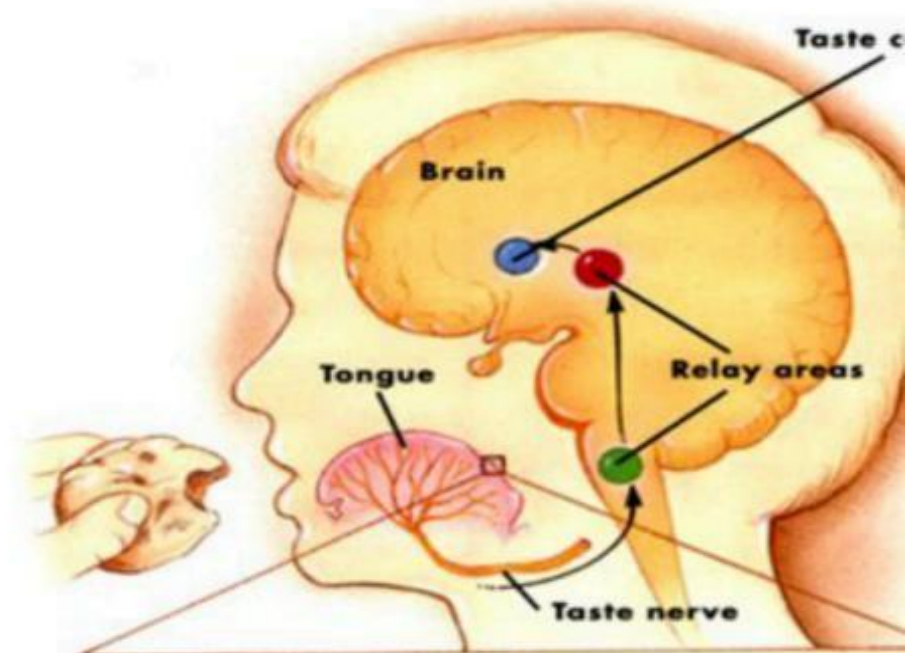
Taste bud



- **Barrel- shaped bodies** found in the oral epithelium.
Found in tongue papillae except filiform, soft palate, epiglottis, larynx & pharynx.

Formed of Supporting/ sustentacular cells at the periphery.
10-14 slender, darkly stained **Taste cells or neuro- epithelial cells** contain villi that project into the taste pore & are related at the base with nerve ending coming from **chorda tympani**

- **Basal cells** with high turn over

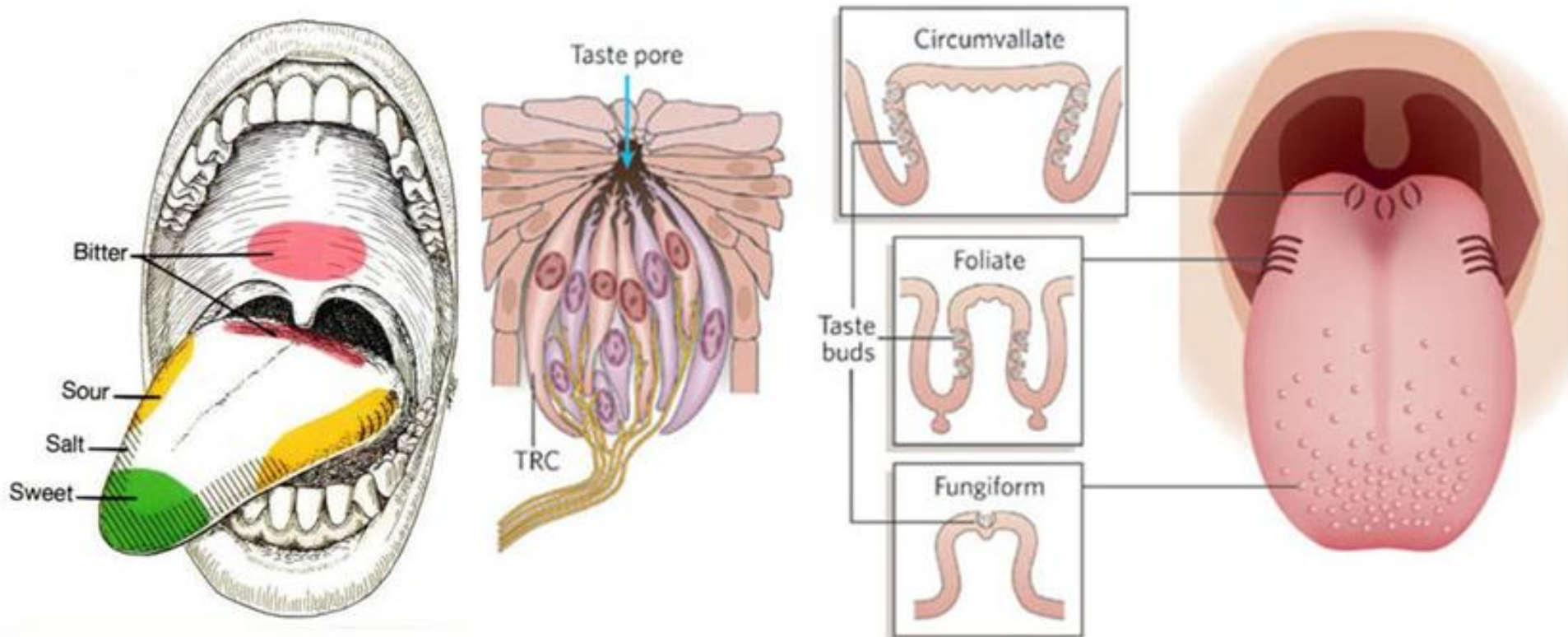


TASTE SENSATIONS



The primary taste sensation are perceived in the following regions

	Sweet	Salt	Bitter	Sour
Tasted best at	Tip of tongue	Lateral borders of tongue body	Posterior part in the middle area	Posterior part in the lateral
Papillae	Fungiform	Fungiform	• Circumvallate • soft palate	• Foliate • Soft palate



Applied Aspects

Tongue Tie: Frenulum extends towards the tip of the tongue.

Clinical Features-

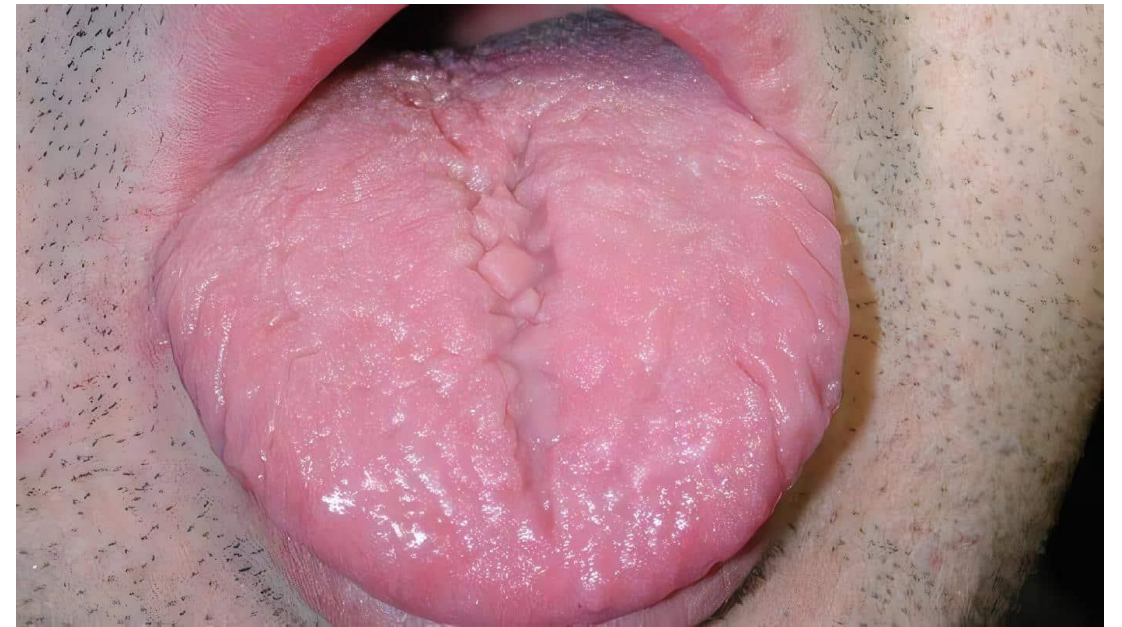
- Inhibited movements of tongue.
- May interfere with normal speech.

Sublingual route of oral medication-

- Some lipid soluble drugs can diffuse through the thin mucous membrane of sublingual region of oral cavity. e.g. Nitroglycerine (Sorbitrate).



Microglosia and macroglosia



THANK YOU